CSE 3967: Machine Learning Concepts 1

	ITER, SIKSHA 'O' ANUSANDHAN **(Deemed to be University)**		**LESSON PLAN**
Programme	**B.Tech.**	Academic Year	**2025-26**
Department	**CSE**	Semester	5^{th}
Credit	**2**	Grading Pattern	**5**
Subject Code	**CSE 3967**		
Subject Name	**Machine Learning Concepts 1**		
Weekly Course Format	**0L-4P**		
Subject Coordinator (s)	**Dr. Sutapa Mahato**		

Text Books(s):

(1) Machine Learning by Dutt, Pearson India
(2) Machine Learning with Python for Everyone by Fenner, Pearson India
(3) Practical Handbook of Machine Learning by Bhattacharyya, Gk Publications

Course Outcomes	Students will be able to	
	CO1	Understand about basic concepts, limitations and applications at Machine Learning and implement these through Numpy library at Python programming.
	CO2	Utilize the Matplotlib at Python programming to understand the different data types, issues and remediation at data and data pre-processing.
	CO3	Apply Python programming for Model selection, performance evaluation and improvement of the Model.
	CO4	Understand about the construction, extraction and selection at different and also can apply concepts at Python programming for these.
	CO5	Utilize the basic concepts of probability as well as Bayesian concept learning in context of Machine Learning and implement them through Python programming.
	CO6	Develop Python applications to analyse and predict outcomes from dataset by using Regressing and Classification techniques.

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Sl.No.	Lessons/Topics to be covered	Book Reference (sections)	Mapping with COs	Home Work/ Assignments/ Quizzes
1	**Introduction to Machine Learning:** Human learning and it's types, Machine learning and it's types, Well-posed learning problem	**Dutt:** Ch-1:1.1-1.4	**CO1**	
2	Applicatiions of machine learning, Issues in machine learning	**Dutt:** Ch-1:1.5-1.7, **SB-**Ch-1	**CO1**	
3	**Preparing to Model:** Basic data types, Exploring numerical data	**Dutt:** Ch:2-2.1-2.3, **SB:** C,D,E.6	**CO2**	
4	Exploring categorical data, Exploring relationship between variables	**Dutt:** Ch:2-2.4, **SB:** C,D,E.6	**CO2**	
5	Data issues and remediation	**Dutt:** Ch:2-2.5, **SB:** C,D,E.6	**CO2**	
6	Data pre-processing	**Dutt:** Ch:2-2.6, **SB:** C,D,E.6	**CO2**	**Assignment-1, Quiz-1**
7	**Modelling and Evaluation:** Selecting a model, Training model-Holdout, k-fold cross-validation, bootstrap sampling	**Dutt:** Ch:3-3.1-3.3, **SB:** Ch(s):2-6,8,12,15 **MF:** Ch(s):4,5,6	**CO3**	
8	Model representation and interpretability, under-fitting, over-fitting, bias-variance tradeoff	**Dutt:** Ch:3-3.4, **SB:** Ch(s):2-6,8,12,15, **MF:** Ch(s):4,5,6	**CO3**	
9	Model performance evaluation-Classification, regression, clustering	**Dutt:** Ch:3-3.5, **SB:** Ch(s):2-6,8,12,15, **MF:** Ch(s):4,5,6	**CO3**	
10	Performance improvement	**Dutt:** Ch:3-3.6, **SB:** Ch(s):2-6,8,12,15, **MF:** Ch(s):4,5,6	**CO3**	
11	**Basics of Feature Engineering:** construction, Feature extraction	**Dutt:** Ch:4-4.1,4.2, **SB:** Ch(s):3,10, **MF:** Ch(s):9,10	**CO4**	

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12	Feature selection	**Dutt:** Ch:4-4.3, **SB:** Ch(s):3,10, **MF:** Ch(s):9,10	**CO4**	**Assignment-2, Quiz-2**
13	**Brief Overview of Probability:** Basic concept of probability, Random variables	**Dutt:** Ch:5-5.1-5.4, **SB:** E-1,E-9, **MF:** 1.12	**CO5**	
14	Discrete distributions-Binomial, Poisson, Bernoulli etc.	**Dutt:** Ch:5-5.5, **SB:** E-1,E-9, **MF:** 1.12	**CO5**	
15	Continuous distribution-Uniform, Normal, Laplace	**Dutt:** Ch:5-5.6,5.7, **SB:** E-1,E-9, **MF:** 1.12	**CO5**	
16	Central theorem, Monte Carlo approximation	**Dutt:** Ch:5-5.8-5.11, **SB:** E-1,E-9, **MF:** 1.12	**CO5**	
17	**Bayesian Concept Learning:** Bayes theorem-Prior and posterior probability, likelihood	**Dutt:** Ch:6-6.1-6.3, **MF:** 2.6	**CO5**	
18	Concept learning, Bayesian Belief Network	**Dutt:** Ch:6-6.4,6.5, **MF:** 2.6	**CO5**	**Assignment-3, Quiz-3**
19	**Supervised Learning:** Basics of supervised learning-classification	**Dutt:** Ch:7-7.1-7.3, **SB:** Ch(s): 7,11,13 **MF:** Ch(s):2,7	**CO6**	
20	k-Nearest Neighbour	**Dutt:** Ch:7-7.4,7.5.1, **SB:** Ch(s): 7,11,13 **MF:** Ch(s):2,7	**CO6**	
21	Decision tree, Random Forest	**Dutt:** Ch:7-7.5.2,7.5.3, **SB:** Ch(s): 7,11,13 **MF:** Ch(s):2,7	**CO6**	

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22	Support vector-machine	**Dutt:** Ch:7-7.5.4, **SB:** Ch(s): 7,11,13 **MF:** Ch(s):2,7	**CO6**	
23	**Supervised Learning:Regression:** Simple linear regression	**Dutt:** Ch:8-8.1,8.2,8.3.1-8.3.3, **SB:** Ch: 2 **MF:** 3.3,7.4	**CO6**	
24	Other regression techniques	**Dutt:** Ch:8-8.3.3-8.3.8, **SB:** Ch: 2 **MF:** 3.3,7.4	**CO6**	**Assignment-4, Quiz-4**

List of Assignments:

- Assignment-1: Introduction to Machine Learning, Preparing to Model
- Assignment-2: Modelling and Evaluation, Basics of Feature Engineering
- Assignment-3: Brief Overview of Probability, Bayesian Concept Learning
- Assignment-4: Supervised Learning:Classification, Supervised Learning:Regression